Section 2: Multivariate Distributions

Let's begin with two random variables and let $\mathcal{C}$ = {results of 3 coin tosses}:

$$\text{\# outcomes} = \{H \cup T\}^3 = 8$$

Let's now define the following random variables represented by C_i, i.e. the result from coin i:

$$X_1 = \{\#H \in C_1, C_2\}$$
$$X_2 = \{\#H\}$$

We can clearly see that there must be some degree of overlap between these two random variables:

	C_1	C_2	C_3	(x_1, x_2)
$\mathcal{C} =$	T	T	T	(0, 0)
	T	T	H	(0, 1)
	T	H	T	(1, 1)
	H	T	T	(1, 1)
	T	H	H	(1, 2)
	H	T	H	(1, 2)
	H	H	T	(2, 2)
	H	H	H	(2, 3)

So, if we are interested in (X_1, X_2), i.e. an ordered pair, then we are no longer interested in $\mathcal{C}$ but now $\mathcal{D}$, a space of ordered pairs, where

$$\mathcal{D} = \{\overset{C_1}{(0,0)}, \overset{C_2}{(0,1)}, (1,1), (1,2), (2,2), (2,3)\}$$

X_1 and X_2 are two random variables defined on space $\mathcal{C}$ while (X_1, X_2) is a vector from $\mathcal{C}$ to $\mathcal{D}$, thus (X_1, X_2) is a **random vector**

Question: Why do you think multivariate Distributions are important?

A: Think about correlation from STAT 231 and how X and Y are ordered pairs that allow us to determine if there is some association between X, Y

Definition 2.1.1: Given a random experiment with $\mathcal{C}$, consider two random variables X_1 and X_2 that assign to each element c of $\mathcal{C}$ **one and only one** ordered pair of numbers $X_1(c) = x_1$ and $X_2(c) = x_2 \ni$

$$\exists\, x_1, x_2 \in \mathcal{D} \ni c \in \mathcal{C}: X_1(c) = x_1, X_2(c) = x_2$$

then we say (X_1, X_2) is a **random vector** (R$\overline{\text{V}}$) and the space of (X_1, X_2) is a set of ordered pairs $\mathcal{D} = \{(x_1, x_2): x_1 = X_1(c), x_2 = X_2(c)\}$

Notation:

$\mathbb{X} = \vec{X} = (X_1, X_2)$ } random vector

Let $A \subset \mathcal{D}$, then similar to a random variable, we can define the probability of A as $P_{X_1,X_2}(A)$ through the CDF in a similar fashion as before

$$F_{X_1,X_2}(x_1, x_2) = P(X_1 \leq x_1, X_2 \leq x_2)$$

$\forall\ (x_1, x_2) \in \mathbb{R}^2$ and, importantly, we can define any joint interval via a similar method

$$P(a < X_1 < b, c < X_2 < d) = F(b, d) - F(a, d) - F(b, c) + F(a, c)$$

NOTE: We talked about solving any probability (and joint probability) based on the CDF in STAT333 and you should review that proof/prove to yourself the above CDF works.

Example 2.1: Consider the previously discussed discrete R$\overline{V}$

$$\mathcal{C} = \begin{array}{ccc} \multicolumn{3}{c}{\overbrace{\qquad\qquad}^{X_1}} \\ \mathrm{T} & \mathrm{T} & \mathrm{T} \\ \mathrm{T} & \mathrm{T} & \mathrm{H} \\ \mathrm{T} & \mathrm{H} & \mathrm{T} \\ \mathrm{H} & \mathrm{T} & \mathrm{T} \\ \mathrm{T} & \mathrm{H} & \mathrm{H} \\ \mathrm{H} & \mathrm{T} & \mathrm{H} \\ \mathrm{H} & \mathrm{H} & \mathrm{T} \\ \mathrm{H} & \mathrm{H} & \mathrm{H} \\ \multicolumn{3}{c}{\underbrace{\qquad\qquad}_{X_2}} \end{array}$$

		X_2			
		0	1	2	3
X_1	0	1/8	1/8	0	0
	1	0	2/8	2/8	0
	2	0	0	1/8	1/8

The **support** are all the points $(x_1, x_2) \ni p(x_1, x_2) > 0$, thus, although $X_1 \in (0, 1, 2), X_2 \in (0, 1, 2, 3)$, not all pairings $(x_1, x_2) \in \mathcal{D}$. What we mean by that is we are interested in pairs with non-zero probabilities, i.e.

$$S_{X_1, X_2} \in \{(0,0), (0,1), (1,1), (1,2), (2,2), (2,3)\}$$

As such, the following pairs are not part of the support of $\mathcal{S}_{X_1,X_2}$

$$S_{X_1, X_2} \notin \{(0,2), (0,3), (1,0), (1,3), (2,0), (2,1)\}$$

Now for something a little more challenging, a continuous random vector. We say that (X_1, X_2) is a continuous random vector with $\mathcal{D}$ if its CDF $F(x_1, x_2)$ is continuous, thus

$$F(x_1, x_2) = \int\int f(x_1, x_2)\, dx_1\, dx_2 \quad \forall\ (x_1, x_2) \in \mathbb{R}^2$$

and

$$\frac{\partial}{\partial x_1 \partial x_2} F(x_1, x_2) = f(x_1, x_2)$$

Example 2.2: Let X_1 and X_2 be defined by the joint PDF:

$$f(x_1, x_2) = \begin{cases} 6 \cdot x_1^2 \cdot x_2, & 0 < x_1, x_2 < 1 \\ 0, & \text{e/w} \end{cases}$$

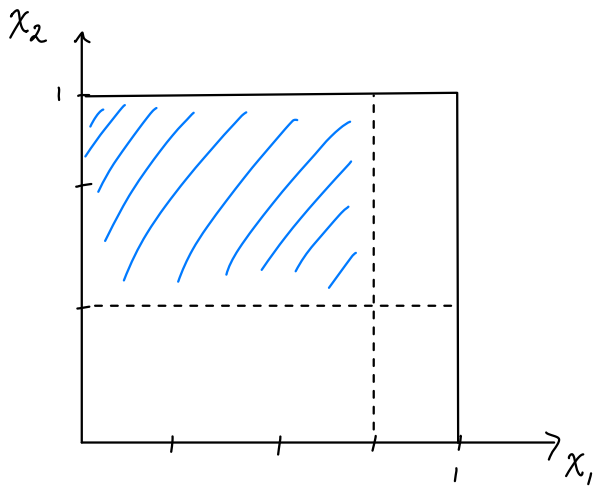
Then if we were interested in the probability of $P\left(0 < X_1 < \frac{3}{4}, \frac{1}{3} < X_2 < 2\right)$:

$$\begin{aligned} Pr\left(0 < x_1 < \tfrac{3}{4},\ \tfrac{1}{3} < x_2 < 2\right) &= \int_0^{\frac{3}{4}}\int_{1/3}^{2} 6x_1^2 x_2\, dx_2\, dx_1 \\ &= \int_0^{\frac{3}{4}} 3x_1^2 \int_{1/3}^{2} 2x_2^2\, dx_2\, dx_1 \\ &= \int_0^{\frac{3}{4}} 3x_1^2 \left(x_2^2 \Big|_{\frac{1}{3}}^{2}\right) dx_1 \\ &= \int_0^{\frac{3}{4}} 3x_1^2 \left(4 - \tfrac{1}{9}\right) dx_1 \\ &= \tfrac{35}{36}\left(\tfrac{27}{64}\right) \end{aligned}$$

the support for X_2 is limited to $0 < x < 1$

$\Rightarrow Pr\left(0 < x_1 < \frac{3}{4},\ \frac{1}{3} < x_2 < 2\right) = Pr\left(0 < x_1 < \frac{3}{4},\ \frac{1}{3} < x_2 < 1\right) = \frac{3}{8}$

For a continuous random vector, the support **contains all points (x_1, x_2) for which $f(x_1, x_2) > 0$**, thus if we were to draw out the support, the probability is the volume under the surface $f(x_1, x_2)$ and above the following rectangular set:



Reminder about notation:

$p(x,y)$, $p(x_1, x_2)$ or any variable will represent a discrete RV (i.e. PMF)
while $f(x,y)$, $f(x_1, x_2)$, or any variable will represent a continuous RV (i.e. PDF)
while the upper case F represents the CDF of our random variable

Let (X_1, X_2) be a R$\overline{\text{V}}$, then X_1 and X_2 are random variables and we can obtain their respective distributions via their joint distributions. For discrete variables,

$$F(x_1) = P(X_1 \leq x_1, -\infty < x_2 < \infty) \quad \forall\, x_1 \in \mathbb{R}$$

Recall, we are going to (essentially) add up all of X_2 over its support to "remove" it so that we can obtain our marginal distribution for X_1; that is

$$F(x_1) = \sum \sum_{\mathcal{D}_{X_2}} p(x_1, x_2) = \sum_{x_1} p(x_1)$$

Example 2.3: Throwing back to STAT231, if you took my class I would have shown an example showing blood type and Rh factor:

	A	B	AB	O
$+$	0.33	0.09	0.03	0.37
$-$	0.07	0.02	0.01	0.08

However, the above is a joint distribution and if we wanted to calculate (for example) the probability that an individual has blood type AB, then we can address this intuitively or via the steps above by adding up over all values of X_2 (Rh factor):

X_2 = RH

$$Pr(AB) = \sum_{D_{X_2}} p(x_1, x_2) = \sum_{D_{X_2}} p(x_1 = AB, x_2)$$
$$= p(AB, (+)) + p(AB, (-))$$
$$= 0.03 + 0.01$$
$$= 0.04$$

Example 2.4: Going back to the first discrete example, we established that the distribution could be presented as:

		X_2				
		0	1	2	3	$Pr(X_1)$
	0	1/8	1/8	0	0	2/8
X_1	1	0	2/8	2/8	0	4/8
	2	0	0	1/8	1/8	2/8
$Pr(X_2)$		1/8	3/8	3/8	1/8	1

$$Pr(X_2 = 0) = \sum_{D_{X_1}} Pr(x_1, 0)$$
$$= Pr(0,0) + Pr(1,0) + Pr(2,0)$$
$$= 1/8 + 0 + 0$$
$$= 1/8$$

Similarly, for a continuous random variable, we can take the integral of the other random variable to determine the marginal PDF of the random variable of interest:

$$f(x_1) = \int_{-\infty}^{\infty} f(x_1, x_2)\ dx_2 \qquad \forall\, x_1 \in \mathcal{D}_{X_1}$$

Example 2.5: Let X_1 and X_2 be defined by the joint PDF

$$f(x_1, x_2) = \begin{cases} x_1 + x_2, & 0 < x_1, x_2 < 1 \\ 0, & \text{e/w} \end{cases}$$

$$f(x_2) = \int_{-\infty}^{\infty} f(x_1, x_2)\ dx$$

$$= \int_0^1 x_1 + x_2\ dx_1$$

$$= x_2 + \frac{1}{2}$$

$$f(x_1) = x_1 + \frac{1}{2}$$

$$Pr\left(X_1 \le \frac{1}{2}\right) = Pr\left(X_1 \le \frac{1}{2},\ 0 < x_2 < 1\right)$$

$$= \int_0^{1/2} x_1 + \frac{1}{2}\ dx_1$$

$$= \left(\frac{1}{2}x_1^2 + \frac{1}{2}x_1\right)\Big|_0^{\frac{1}{2}}$$

$$= \left[\frac{1}{2}\left(\frac{1}{2}\right)^2 + \frac{1}{2}\left(\frac{1}{2}\right)\right] - \left[\frac{1}{2}(0)^2 + \frac{1}{2}(0)\right]$$

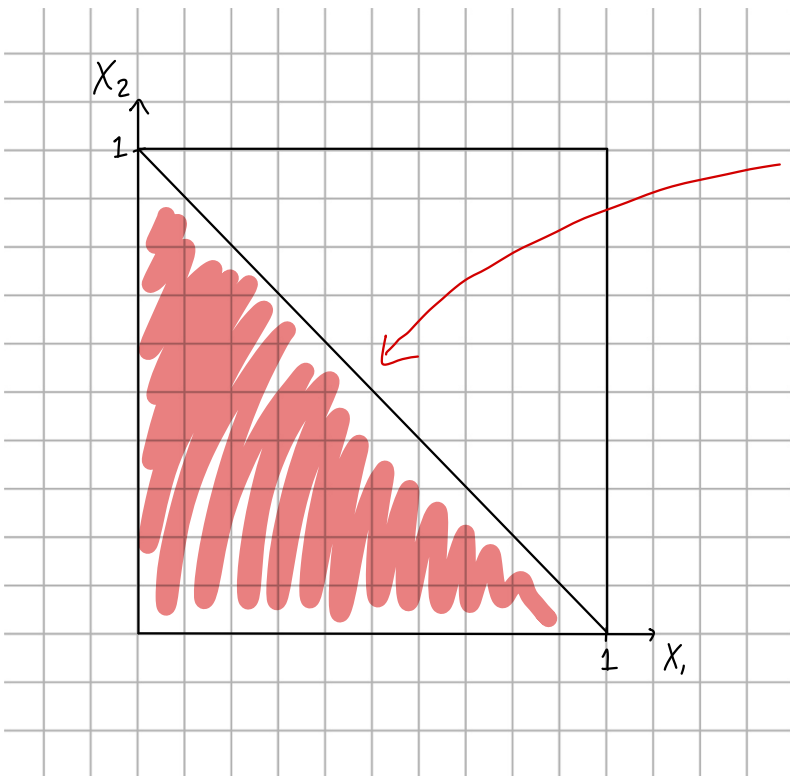
$$= \frac{1}{8} + \frac{1}{4}$$

$$= \frac{3}{8}$$

Example 2.6: Consider the same joint distribution above; however, with the following probability of interest: $P(X_1 + X_2 \leq 1)$. In this example, **the integrand of X_1 (or X_2) is a dependent on X_2 (or X_1),** i.e. it is a function of the other random variable:

$X_1 \leq 1 - X_2$ <u>or</u> $X_2 \leq 1 - X_1$

As such, the support is no longer defined where we can view X_1 and X_2 "independently",



$X_2 \leq 1 - X_1$

$\{(x_1, x_2) : 0 < x_1,\ x_1 + x_2 < 1\}$

$Pr(X_1 + X_2 \leq 1)$

$= Pr(X_2 \leq 1 - X_1)$

$= \int_0^1 \int_0^{1-x_1} x_1 + x_2 \, dx_2 \, dx_1$

$= \int_0^1 \left(x_1 x_2 + \frac{1}{2} x_2^2\right)\Big|_0^{1-x_1} dx_1$

$\vdots$

$= \left(\frac{1}{2} x_1^2 - \frac{1}{3} x_1^3 + \frac{1}{2} x_1 - \frac{1}{2} x_1^2 + \frac{1}{6} x_1^3\right)\Big|_0^1$

$= \frac{1}{3}$